

Understanding Suicide Patterns: A Multi-Level Analysis of Global, East Asian, and Hong Kong

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Abstract

This report investigates the alarming issue of high suicide rates in East Asian countries, particularly China, Japan, and South Korea, recognizing it as a critical indicator of societal mental health and psychological stress. The study addresses the need for in-depth research into the correlation between economic fluctuations and suicide rates. Existing approaches often examine macroeconomic indicators and suicide rates using time-series analysis methods like the Augmented Dickey-Fuller (ADF) test, Vector Autoregression (VAR) models, and Granger causality tests. While these methods can reveal interdependencies and predictive power, they may not fully capture the nuances of localized risk factors or individual-level influences. To overcome these limitations, this study complements macroeconomic analysis with unsupervised clustering algorithms and association rule mining on detailed, individual-level data from Hong Kong. This mixed-methods approach intuitively allows for the identification of high-frequency suicide hotspots based on geographical data and uncovers patterns between suicide locations and methods, as well as correlations among age, gender, and reasons for suicide. The empirical results, such as the clustering of suicide incidents in high-population-density urban areas and the strong association between certain locations and methods, justify this integrated approach, providing a more scientific foundation for targeted suicide prevention strategies and social policies. Globally, factors like life expectancy and health spending show complex associations with suicide rates, while in East Asia, particularly South Korea, GDP fluctuations appear linked to suicide rates in certain age groups, though the impact of unemployment is less clear; in Japan, short-term economic factors show less direct impact, but historical crises highlight a link to economic stress. Analysis of Hong Kong data reveals suicide hotspots in high-density urban areas and associations between specific locations and methods, as well as correlations between age, gender, and reasons for suicide at the individual level.

1 Introduction

1.1 Background

Suicide has become a major global public health challenge that urgently requires attention. According to statistics from the World Health Organization, more than 700,000 people die by suicide each year, with one person dying every 40 seconds on average [World Health Organization \(2021\)](#). This number continues to rise amid ongoing social transformation and increasing mental health pressures. Due to disparities in economic development levels, cultural values, social support systems, and mental health services, suicide rates vary significantly across countries, regions, and demographic groups. East Asian countries—namely China, Japan, and South Korea—have long maintained suicide rates above the global average. Meanwhile, Hong Kong, as a highly urbanized international metropolis, has seen a steady increase in suicides in recent years, largely attributed to rising living costs and intense societal pressures.

Although previous studies have explored suicide risk factors from psychological, sociological, and economic perspectives—such as individual emotional and cognitive mechanisms, family and social support networks, unemployment rates, and income inequality—there remains a lack of comprehensive analysis capturing the multi-level and multi-dimensional interactions among these factors. In particular, there is a gap in constructing an integrated framework that connects global, regional, and local perspectives. Furthermore, limited research has attempted to combine traditional statistical methods with advanced techniques such as machine learning, time-series modeling, clustering, and association rule mining to uncover hidden patterns and potential causal relationships.

1.2 Research Motivation

In the face of the complex nature of suicide behaviors, our study is driven by two primary motivations:

1. Lack of data-driven global perspective: Most existing studies focus on a single country or region, failing to capture the spatiotemporal evolution of suicide rates and their underlying driving factors across multiple scales [Franklin et al. \(2017\)](#). By conducting comparative analyses across geographic regions and levels, we aim to identify both universal patterns and localized variations, providing integrated insights for prevention strategies that encompass diagnosis, forecasting, and intervention.
2. Need for multi-method deep exploration: Suicide is influenced by a multitude of economic, sociocultural, and psychological factors that interact in nonlinear and complex ways. Traditional single-model approaches often fall short in uncovering these underlying associations. Therefore, we seek to integrate various analytical techniques—including machine learning, time series analysis, clustering, and association rule mining—to predict suicide rate trends, classify population-level differences, and identify potential influencing factors.

1.3 Research Significance

This study carries significant theoretical and practical implications. By constructing a three-tier analytical framework—encompassing global, regional, and individual levels—it enriches the theoretical foundation of multi-scale and multi-method integration in suicide research, and provides a transferable paradigm for big data applications in public health. Through the combined application of machine learning and time series modeling, the study explores hybrid models suited for high-dimensional behavioral data, contributing to enhanced precision and interpretability in mental health data analysis.

The research findings can offer empirical support for mental health interventions, public policy formulation, and resource allocation by the World Health Organization, East Asian governments, and the Hong Kong administration. Furthermore, through case studies in East Asia and Hong Kong, the study identifies high-risk demographic groups, aiding in the design of targeted and differentiated prevention strategies aimed at reducing suicide rates and improving mental health service systems.

1.4 Our Work

This study integrates suicide rate and socioeconomic indicator data from 160 countries globally, as well as from three East Asian countries (China, Japan, and South Korea) and Hong Kong. After data cleaning and standardization, we apply a combination of techniques—regression analysis, time-series forecasting, vector autoregressive causal inference, spatial and individual-level clustering, and association rule mining—to construct and validate a three-level analytical framework. Finally, we propose actionable policy recommendations based on insights from regional case studies.

2 Analysis of the World Suicide Rate

2.1 Data Sources and Indicator Selection

This study analyzes panel data from 160 countries (2020-2022), focusing on 13 socioeconomic and health indicators to explore factors driving national suicide rates. Key variables include life expectancy (health status), per capita GDP and unemployment rate (economic prosperity and labor pressure), Gini index and poverty ratio (psychological stress from inequality and poverty), health spending and UHC (medical access), HDI (development across education, income, and health), tertiary school enrollment (education level), crime rate (public safety), and population and PM2.5 concentration (control for scale and environmental stress). Suicide rate data is sourced from the kaggle as the dependent variable [Tavoosi \(2022\)](#). Other related data are sourced from the WHO [World Health Organization \(2023\)](#).

2.2 Data Preprocessing

To minimize the impact of short-term fluctuations, data for all indicators was averaged over 2020-2022. Countries with over 50% missing values in key indicators were excluded. For occasional missing values, median imputation was applied based on the most recent records (e.g., 2020-2022 median for missing GDP in 2021). Continuous variables were standardized using Z-scores, and Variance Inflation Factors (VIFs) assessed multicollinearity. After preprocessing, the dataset covers 83% of the global population,

reducing missing values from 12% to 3.5%. A comparison of descriptive statistics confirms the stability of data distribution, ensuring reliable analysis.

2.3 Regression Analysis and Results Interpretation

Based on the preprocessed dataset, both linear regression and random forest regression models were applied to explore the driving factors of suicide rates globally.

Linear Regression Analysis As shown in Table 1 (n=160), the linear model demonstrated moderate explanatory power (Adjusted $R^2 = 0.228$, $F = 10.395$, $p < 0.001$). Life expectancy exhibited a significant negative association with suicide rate (Beta = -0.994, $p < 0.001$), indicating that countries with higher health standards and public health infrastructure tend to have lower suicide rates. Longer life expectancy is typically associated with greater access to crisis intervention and medical services, which mitigates suicidal behaviors. Additionally, improved welfare and life quality elevate overall well-being and reduce suicide tendencies. PM2.5 concentration showed a significant negative association with suicide rates (Beta

Table 1: Linear Regression Analysis Results (n=160)

Variable	B	Std. Error	Beta	t	P	VIF	R^2	Adj. R^2	F-statistic
Constant	9.704	0.455	—	21.319	0.000***	—			
Life expectancy	-6.514	1.095	-0.994	-5.948	0.000***	5.753			
PM2.5 concentration	-1.486	0.520	-0.227	-2.856	0.005***	1.299	0.252	0.228	F=10.395, P=0.000***
per_GDP	1.331	0.629	0.203	2.117	0.036**	1.895			
UHC_index	1.827	0.998	0.279	1.831	0.069*	4.775			
health_spending	2.075	0.701	0.317	2.961	0.004***	2.355			

= -0.227, $p = 0.005$), which may reflect the coexistence of economic growth and better healthcare access in high-pollution areas [Go et al. \(2024\)](#). However, collinearity with economic indicators may confound the results, suggesting the need for further control in future studies.

Per capita GDP (Beta = 0.203, $p = 0.036$) and health spending (Beta = 0.317, $p = 0.004$) were positively associated with suicide rates, indicating that economic growth and healthcare investment may increase societal pressure and strain rather than reduce risk. Health spending, possibly a response to high suicide rates, may have limited preventive effects without equitable mental health coverage.

UHC was marginally significant (Beta = 0.279, $p = 0.069$), suggesting complex, possibly nonlinear relationships. All independent variables had VIF values below 6, indicating no multicollinearity.

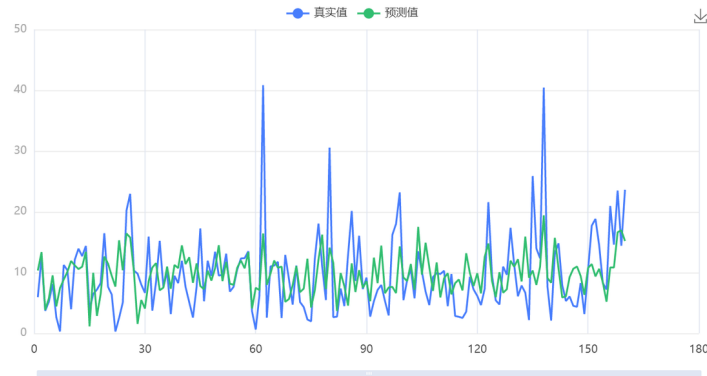


Figure 1: Linear regression prediction graph of suicide rate.

Random Forest Regression Analysis Random forest regression identified the top five factors influencing suicide rates: tertiary education enrollment (26.2%), life expectancy (22.9%), population size (18.8%), health spending (18.1%), and the Gini index (14.0%). These factors suggest that structural elements like education and inequality interact in nonlinear ways with suicide behavior, where higher education may enhance resilience and greater inequality increases mental stress.

PM2.5 and GDP were not among the top five, indicating their effects may be secondary or obscured by complex interactions. Life expectancy's influence was likely linked to demographic and social factors like population density.

While the random forest model performed well on the training set ($R^2 = 0.864$), its generalization capacity was weak: the validation set R^2 was negative (-0.065), and the test set R^2 was only 0.348, suggesting potential overfitting and limited robustness.

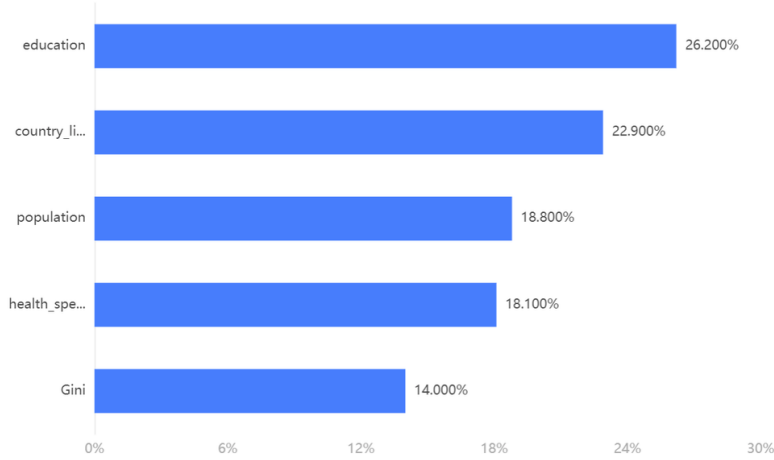


Figure 2: The proportion distribution chart of the five key factors influencing the suicide rate.

Table 2: Random Forest Regression Performance

Dataset	MSE	RMSE	MAE	MAPE (%)	R^2
Training Set	5.275	2.297	1.628	17.648	0.864
Validation Set	37.904	6.018	4.221	43.938	-0.065
Test Set	33.424	5.781	4.407	40.737	0.348

Model Comparison and Policy Implications The comparison of the two models reveals the multidimensional nature of suicide rate determinants. Linear regression highlights direct links between health/environmental factors and suicide, while random forest uncovers the significant role of education, inequality, and demographics through complex interactions. Life expectancy ranked high in both, but its effect was more interdependent in the nonlinear model.

These results suggest that policies should address both public health measures and structural factors—improving education, reducing inequality, and expanding mental health resources. Future research could explore hybrid models, like generalized additive models or multi-task learning, to combine linear and nonlinear factors and validate robustness across variables and analyses.

3 Analysis of Suicide Rate in East Asian Countries

Suicide rate, as a crucial indicator of societal mental health, reflects the psychological stress and mental state of social groups. In recent years, the high suicide rates in East Asia, particularly in China, Japan, and South Korea, have drawn widespread attention. Economic fluctuations, as significant external environmental factors affecting mental health, warrant in-depth research into their correlation with suicide rates and the dynamic mechanisms of change.

This study examines the impact of economic changes on suicide risks across different age groups in the three major East Asian countries, based on time-series data of macroeconomic indicators (GDP per capita and unemployment rate) and suicide rates. The aim is to provide a scientific foundation for suicide prevention and social policies.

3.1 Dataset Introduction

The dataset used in this study was compiled and released by Ding [Ding et al. \(2020\)](#), covering annual suicide rates by gender and age group from 1991 to 2017 across 175 countries, along with macroeconomic indicators: GDP per capita (adjusted for inflation to 2010 USD) and official unemployment rates.

The analysis of East Asia focuses on China, Japan, and South Korea, selected based on their average suicide rates. Suicide data are standardized per 100,000 population, calculated by dividing the number of suicides in a specific age or gender group by the total population of that group and multiplying by 100,000. Both GDP and unemployment figures are derived from official, authoritative statistics, ensuring the scientific validity and completeness of the data.

3.2 Data preprocessing

To ensure data quality, records with anomalies such as negative suicide rates were first removed. Missing values for key economic variables were imputed using each country's historical median. Samples with remaining missing data were then excluded to guarantee data completeness.

Age groups for suicide data were restructured into ten-year intervals (e.g., 10–19 years) to facilitate trend analysis. Suicide rates, GDP per capita, and unemployment rates were aggregated across multiple dimensions—country, year, age group, and gender—using mean or median values to ensure detailed stratification.

Finally, all indicators were standardized using Z-scores to eliminate the influence of different units, ensuring the stability and comparability of subsequent statistical analyses.

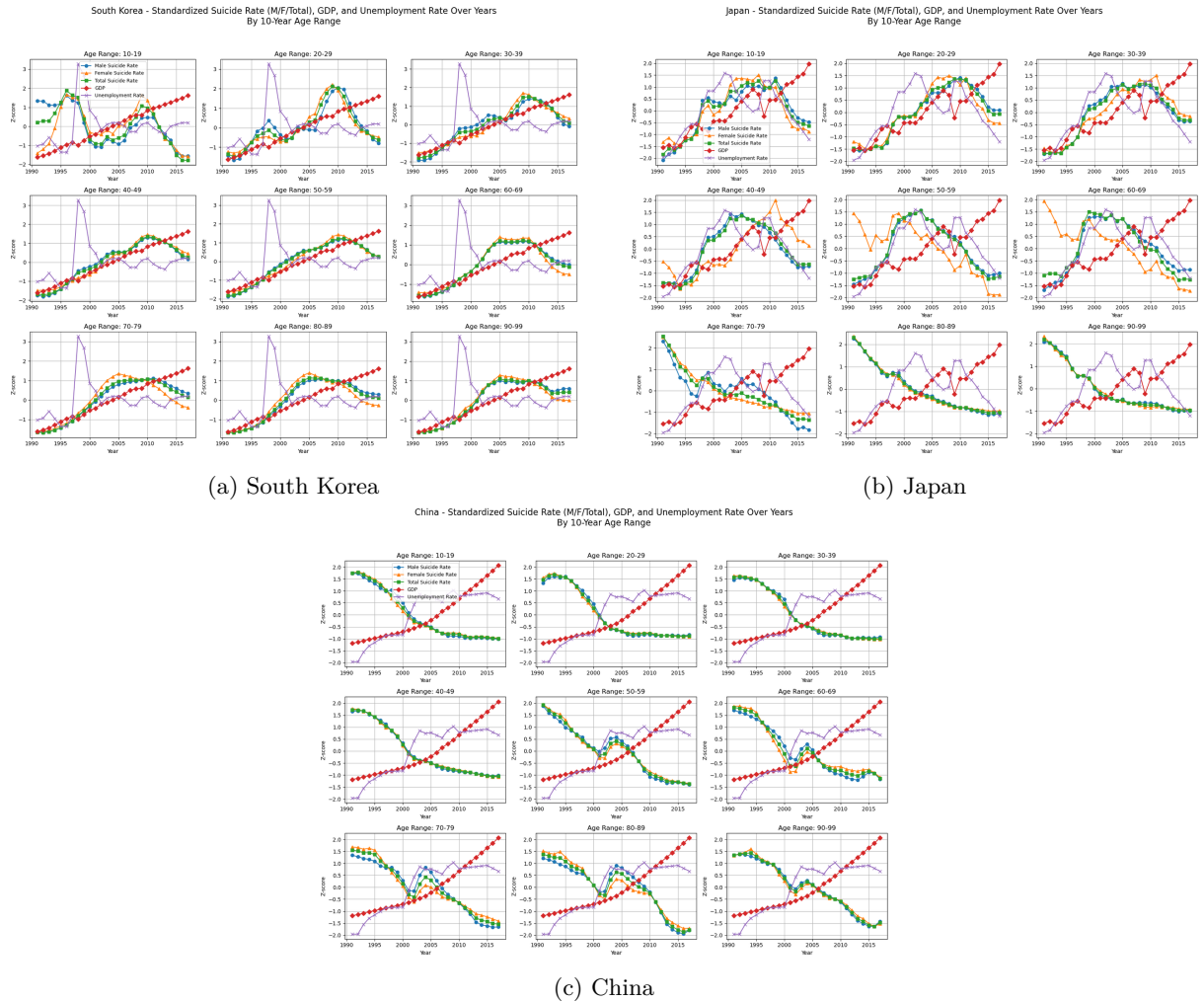


Figure 3: Annual trends of suicide rates, GDP and unemployment rates in different age groups in South Korea, Japan, China.

3.3 Time Series Causality Analysis

In this study, stationarity of time series data is first tested using the Augmented Dickey-Fuller (ADF) unit root test to ensure key statistical properties such as mean and variance remain stable over time.

Stationarity is crucial for valid model estimation and forecasting, while non-stationary series need transformations like differencing to avoid spurious regression, a common issue in economics and public health research .

Next, a Vector Autoregression (VAR) model is employed to analyze interdependencies and time-lagged effects among variables like GDP, unemployment rate, and suicide rate. The VAR model’s ability to treat all variables symmetrically makes it a powerful tool in economic and social science research, allowing the study of how past changes in one variable influence others, which is particularly useful in understanding complex social behaviors such as suicide rates during economic fluctuations. Granger causality tests are then applied to assess whether economic indicators can predict suicide rates, identifying potential causal relationships, a key approach in evidence-based policymaking in economics and public health .

Taira [Taira et al. \(2021\)](#) conducted similar time series analyses on suicide and socioeconomic data in Japan, providing a solid empirical and theoretical framework for this study, which investigates the relationship between suicide rates and macroeconomic factors in East Asia.

3.4 Results

The time series analysis reveals that changes in South Korea’s GDP are notably associated with changes in suicide rates, especially in the 30–39 and 40–49 age groups. During periods of GDP growth, these age groups experience significant changes in suicide rates, as shown by statistically significant VAR coefficients and Granger causality test results ($p < 0.05$). The Granger causality test further indicates that GDP fluctuations can predict suicide rate changes across multiple age groups, suggesting a potential temporal causal relationship. In contrast, the unemployment rate’s impact on suicide rates appears weaker and less stable, showing significance only in the 30–39 age group and minimal influence in other age brackets. Overall, GDP changes exert a stronger influence on suicide rates than unemployment rate fluctuations in South Korea.

Table 3: VAR Model Coefficients and Granger Causality Tests by Age Group (South Korea)

Indicator	Age 20–29	Age 30–39	Age 40–49	Age 50–59
VAR Model Coefficients at Lag 1				
Coefficient of GDP on Rate	1.0222	3.164 (p=0.001, sig.)	1.6371 (p=0.036, sig.)	0.3518 (n.s.)
p-value (GDP → Rate)	0.13	0.001	0.036	0.093
Coefficient of Unemp on Rate	-0.0427	0.1593 (p=0.041, sig.)	0.0764 (n.s.)	-0.0255 (n.s.)
p-value (Unemp → Rate)	0.404	0.041	0.254	0.11
Granger Causality Test p-values				
GDP → Rate	0.0484 (Chi2)	0.015–0.0325 (sig.)	0.0047–0.0841 (borderline sig.)	0.0258–0.0483 (lag 1, sig.)
Unemp → Rate	0.2074 (F)	0.072–0.634 (n.s.)	0.0342–0.4802 (borderline n.s.)	0.0316–0.0561 (lag 1, borderline sig.)

The VAR model reveals that unemployment significantly impacts suicide rates among individuals aged 40–49, a group under high family and career pressure. Granger causality tests show that GDP fluctuations affect suicide rates among those aged 20–49 in the short term, though this influence fades over time, indicating temporary psychological and social instability during economic downturns.

The time-lagged effect of unemployment varies by age, reflecting different responses to economic stress. Combined with Xu Wenbin [Xu et al. \(2017\)](#) , which highlights higher suicide vulnerability among middle-aged groups in China, the findings emphasize the heightened fragility of this demographic under short-term economic volatility.

These results call for targeted mental health and social support measures, particularly for middle-aged adults, to reduce suicide risk linked to economic stress.

Table 4: VAR Model Coefficients at Lag 1 by Age Group (China)

Indicator	20–29	30–39	40–49	50–59
Coef. of GDP on Rate	-0.766 (n.s.)	0.807 (n.s.)	-0.559 (n.s.)	1.701 (n.s.)
p-value (GDP → Rate)	0.739	0.634	0.739	0.369
Coef. of Unemp on Rate	-0.001 (n.s.)	0.160 (n.s.)	0.165 (p=0.049, sig.)	0.164 (n.s.)
p-value (Unemp → Rate)	0.992	0.108	0.049	0.610
Coef. of Unemp on GDP	Non-significant across lags			

VAR model and Granger causality tests in Japan show no significant short-term link between GDP or unemployment fluctuations and overall suicide rates. However, a lagged positive correlation between

Table 5: Granger Causality Test p-values by Age Group (China)

Test	20–29	30–39	40–49	50–59
GDP → Rate (lag 1)	0.0006 (sig.)	0.0012 (sig.)	0.0040 (sig.)	0.998–0.9658 (n.s.)
GDP → Rate (lags 2–5)	0.5356–0.9428 (n.s.)	0.3325–0.8820 (n.s.)	0.1544–0.9450 (n.s.)	–
Unemp → Rate (lags 1–5)	0.1667–0.4543 (n.s.)	0.0060–0.6332 (lag 2 sig.)	0.0016–0.9556 (some sig.)	0.0042–0.2387 (some sig.)

economic changes and suicide among younger groups suggests delayed impacts through employment conditions.

Despite limited short-term statistical evidence, historical data—especially during the 1997–1998 Asian financial crisis—highlight a sharp rise in suicides, notably among middle-aged men, amid economic strain. A spike in 1998 suicide rates supports the view that long-term economic stress plays a key role.

Thus, short-term models may understate the broader socioeconomic influence on suicide, especially across vulnerable demographics.

Table 6: VAR Model Coefficients and Granger Causality Tests by Age Group (Japan)

Indicator	Age 20–29	Age 30–39	Age 40–49	Age 50–59
VAR Model Coefficients at Lag 1				
Coef. of GDP on Rate	-0.1486 (n.s.)	-0.2676 (n.s.)	-0.2595 (n.s.)	-0.0434 (n.s.)
p-value (GDP → Rate)	0.732	0.348	0.614	0.914
Coef. of Unemp on Rate	-0.4153 (n.s.)	-0.3429 (n.s.)	-0.3878 (n.s.)	-0.1303 (n.s.)
p-value (Unemp → Rate)	0.226	0.172	0.385	0.750
Granger Causality Test p-values				
GDP → Rate (lags 1–5)	0.3997–0.7764 (n.s.)	0.1418–0.8086 (mixed)	0.1894–0.9353 (n.s.)	0.1005–0.9353 (mixed)
GDP → Rate (sig. lags)	–	lags 3–5 (² sig.)	–	lag 4 (p=0.0361), lag 5 (p=0.0060)
Unemp → Rate (lags 1–5)	0.9094–0.8482 (n.s.)	0.423–0.8738 (n.s.)	0.1412–0.9662 (n.s.)	0.9256–0.8401 (n.s.)

In summary, the suicide rates in the three East Asian countries show a certain degree of correlation with macroeconomic fluctuations. Although economic factors are not the sole determinants, they play a crucial role in shaping changes in suicide rates through complex time-lag effects and group-specific differences. A deeper understanding of these dynamic relationships can support the development of more targeted and effective suicide prevention social policies.

4 Analysis of Suicide Rate in Hong Kong

This chapter presents an analysis of suicide rates in Hong Kong, employing two key techniques: Suicide Region Clustering and Association Rule Mining.

The first part of the analysis focuses on identifying regions with high suicide rates through clustering methods, providing insights into spatial patterns and the distribution of suicide incidents across districts, sub-districts, and smaller units within the city.

In the second part, association rule mining is applied to explore the relationship between the locations of suicides and the methods used. This method helps uncover patterns and correlations between different factors, offering a deeper understanding of the factors contributing to suicide trends in Hong Kong. The findings aim to support preventive measures and policy recommendations for mental health and suicide prevention strategies.

4.1 Dataset Introduction

Hong Kong Suicide Reports Database [Yeong \(2025\)](#): As of April 11th, it contains a total of 6,341 entries. Updated hourly, this raw dataset compiles suicide cases reported by Hong Kong media, including detailed information such as case ID, timestamp, location (district and coordinates), victim demographics (gender, age, etc.), methods or causes of suicide, and relevant helpline contacts. The database serves as a vital resource for analyzing geographical and social patterns, aiming to inform suicide prevention strategies and public health interventions.

GeoJSON: A JSON-based standard for geospatial data, describing features (e.g., polygons, points) and their properties. Widely used in web maps (e.g., Leaflet) and GIS tools (e.g., QGIS).

4.2 Suicide region clustering

By applying unsupervised clustering algorithms to the geographical data in suicide reports, we can identify high-frequency suicide hotspots and uncover their spatial distribution patterns. Combined with GIS technology, further analysis can reveal environmental characteristics of these clusters (e.g., population density, public facilities, economic conditions), helping to explore potential correlations between suicide incidents and regional social, economic, or environmental factors.

Such analysis provides policymakers with evidence-based insights to optimize mental health resource allocation, enhance preventive measures (e.g., installing barriers, surveillance, or helpline signage) in high-risk areas, and ultimately support targeted suicide prevention strategies.

Data preprocessing During data preprocessing, the latitude and longitude fields were extracted for geospatial analysis. A completeness check revealed only 5 records ($< 0.1\%$ of total data) had missing or unknown coordinates. Given the negligible and random nature of these omissions, the records were excluded to maintain dataset integrity without introducing bias.

Method The workflow begins with data processing: CSV data is loaded with latitude and longitude converted to float, then transformed to radians using `np.radians` for Haversine-based distance calculations.

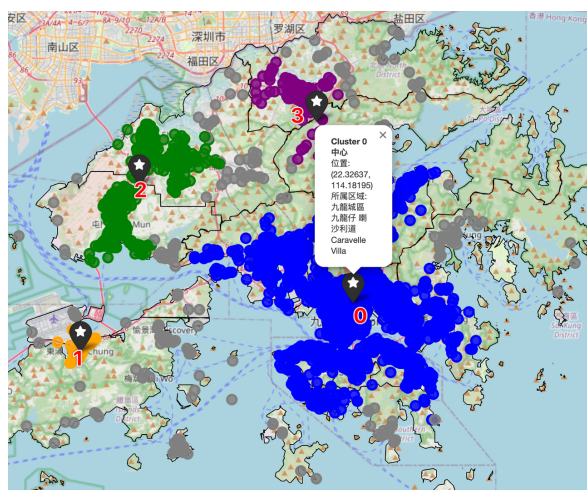
DBSCAN clustering [Ester et al. \(1996\)](#) is applied with a Haversine metric, $\text{eps} = 2 / 6371$ (approx. 2 km), and $\text{min_samples} = 50$.

For visualization, an interactive Folium map is created, overlaying Hong Kong's 18-district GeoJSON and plotting points colored by cluster, with special markers for cluster centers.

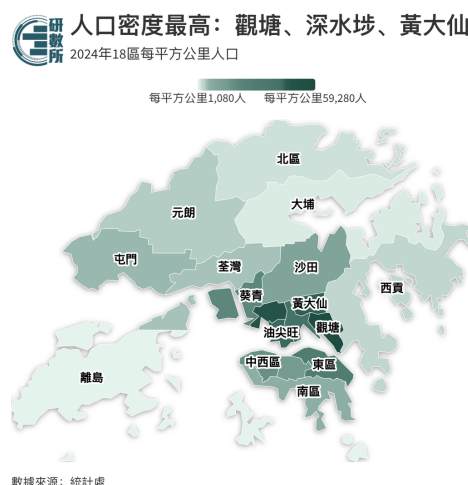
Finally, clustering quality is evaluated using three internal metrics: Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index.

Results DBSCAN Clustering Results:

- Cluster -1 (Noise): 260 points
- Cluster 0: 4,933 points (Major urban clusters, e.g., Kwun Tong, Sham Shui Po, Wong Tai Sin)
- Cluster 1: 821 points (Secondary hotspots, e.g., Sha Tin, Eastern District, parts of Yuen Long)
- Cluster 2: 433 points
- Cluster 3: 84 points



(a) Cluster Results



(b) Population Density in HK

Figure 4: Cluster Results and Population Density in HK

DBSCAN successfully identified four meaningful suicide incident clusters:

- The dominant cluster (Cluster 0) aligns with high-population-density urban areas, suggesting a possible correlation with urbanization stress.
- Smaller clusters (Clusters 1–3) indicate potential localized risk zones, warranting further investigation into geographic, economic, and social factors.
- Noise points (Cluster -1) represent isolated cases or low-density regions, which may still hold valuable insights for outlier analysis.

Evaluation Evaluation Metrics:

- Silhouette Coefficient: 0.5412
- Davies-Bouldin Index: 0.4527
- Calinski-Harabasz Index: 4867.66

The combined metrics confirm stable and meaningful clustering, particularly when accounting for real-world geographic proximity. The spatial distribution aligns with practical living zones, reinforcing the validity of DBSCAN’s density-based approach for this urban context.

4.3 Association Rule Mining

In this section, association rule mining will be performed using the Apriori algorithm [Agrawal et al. \(1993\)](#), based on the features location and method.

Data preprocessing The suicide location descriptors were enhanced by concatenating the `subUnit` (e.g., flat numbers) and `house` (building identifiers) fields. A completeness audit revealed only 7 records (0.05% of total data) had missing values in these fields. Given the negligible impact, these entries were excluded to ensure dataset consistency without introducing bias.

Method The process starts with reading the CSV file and converting each event into a set of items, combining `subUnit` and `House` for location, and including up to three suicide method types (`type1` to `type3`).

These records are then transformed into a one-hot encoded Boolean matrix using `TransactionEncoder`, making them compatible with the Apriori algorithm.

Frequent itemsets are extracted based on a predefined `min_support`, followed by generating association rules with `association_rules`, retaining only those whose consequents are suicide methods.

Finally, rules are sorted by metrics such as support, confidence, and lift, and key patterns are visualized using methods like heatmaps and network graphs.

Results and Evaluation This study reveals the patterns between suicide locations and methods through three key metrics: **Support**, **Confidence**, and **Lift**.

Apriori Results:

1. Near-deterministic Associations

- Open sea locations almost invariably involve drowning (99.3% confidence)
- Tree-related suicides show 100% hanging correlation
- Elevated sites (e.g., rooftops, footbridges) exhibit >95% jumping probability

2. Characteristic Location-Method Pairings

- Vehicle interiors strongly associate with charcoal burning (90.2% confidence)
- Village house units show weaker charcoal burning correlation (47.7% confidence)

Overall, different suicide locations significantly affect the methods used, and this regularity can provide an important basis for formulating targeted intervention measures, such as strengthening protective facilities, restricting access, or increasing monitoring in high-risk locations.

Table 7: Association Rules between Location and Method

No.	Antecedents	Consequents	Support	Confidence	Lift
1	對開海面 _ 非住宅	淹溺	0.070127	0.993492	11.423414
2	一汽車 _ 非住宅	燒炭	0.015465	0.901786	8.462015
3	附近樹上 _ 非住宅	自縊	0.008115	1.000000	6.719136
4	一單位 _ 村屋	燒炭	0.011177	0.477124	4.477152
5	高處 _ 非住宅	跳落	0.084214	0.998185	1.774883
6	一大廈高處 _ 私人屋苑	跳落	0.017915	0.975000	1.733658
7	行人天橋 _ 非住宅	跳落	0.016996	0.948718	1.686925

5 Individual-level Suicide Analysis in Hong Kong

This chapter delves into the individual-level factors influencing suicide rates in Hong Kong [Wong et al. \(2017\)](#)., focusing on Association Rule Mining and Cluster Analysis. The first part uses association rule mining to uncover relationships among key factors such as age, gender, and reason for suicide, shedding light on patterns and correlations that may exist between these variables. The second part of the analysis applies cluster analysis to group individuals based on age, gender, and the primary reason for suicide, aiming to identify distinct profiles within the data. This chapter provides valuable insights into how these personal factors interact and contribute to suicide risk, helping to inform targeted interventions and policies to address mental health challenges.

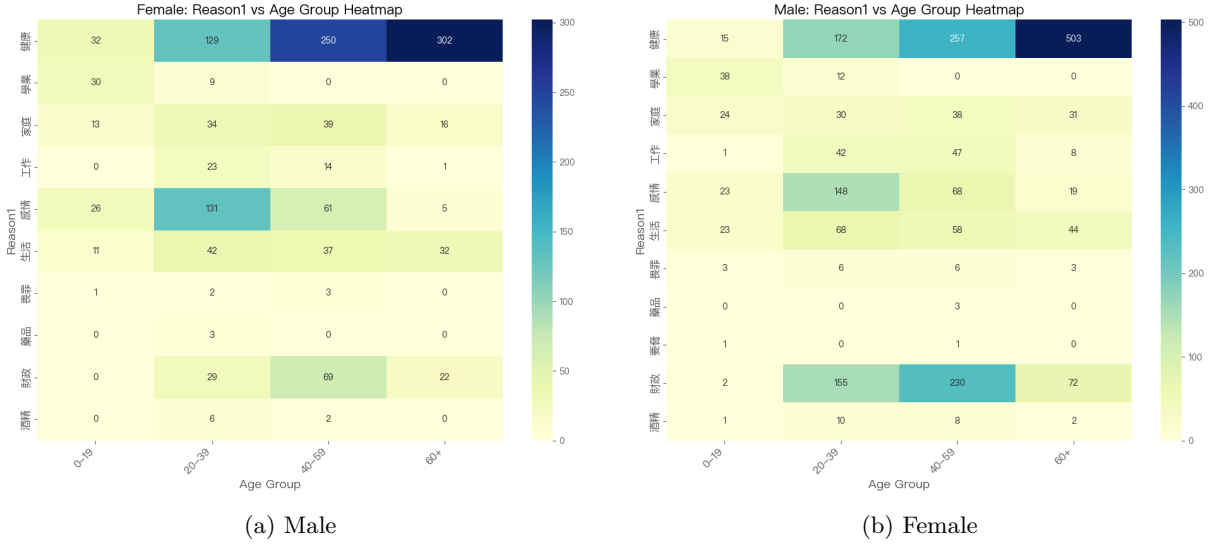


Figure 5: Heatmap of Different Age Groups and Causes of Death

5.1 Cluster Analysis

This dataset investigates suicide cases in Hong Kong with recorded causes of death from 2019 to the present.

In this section, clustering analysis will be conducted using K-Prototypes based on the features age, gender, and reason.

Data Preprocessing Before conducting cluster analysis, we performed the following preprocessing steps on raw data to ensure its quality and suitability for analysis:

- **Feature Selection.** Selected the following three variables as input features for cluster analysis:
 - Age: Represents the individual’s age
 - Gender: Represents gender
 - reason1: Indicates the primary reason behind the individual’s behavior (e.g., seeking help, making decisions)

- **Data Cleaning**

- Removed records where Age or reason1 was marked as “Unknown” to eliminate missing or invalid information.
- Checked the format of Age, excluded non-numeric records, and converted it into a numerical type.

- **Categorical Encoding**

- Used LabelEncoder to convert Gender and reason1 into numerical codes so they could be processed by the K-Prototypes algorithm.
- Maintained consistent mappings of encoded categorical variables to preserve interpretability during clustering.

- **Identify Categorical Feature Indices**

- To inform the model which columns are categorical during clustering, the indices of the encoded categorical features were recorded for use with the `categorical` parameter in KPrototypes.

Method K-Prototypes is a clustering algorithm designed for mixed data types—that is datasets containing both numerical and categorical variables. It combines the strengths of K-Means and K-Modes, with its core ideas as follows:

- **Numerical Features:** Similarity between samples is measured using Euclidean distance.
- **Categorical Features:** Uses simple matching dissimilarity—a distance of 1 if categories differ, and 0 if they are the same.
- **Distance Weighting Parameter (γ , Gamma):** Balances the influence of numerical and categorical variables in the distance calculation. The value of γ is typically set empirically or optimized through parameter tuning.

The loss function of K-Prototypes is defined as follows:

$$\sum_{i=1}^n \sum_{j=1}^k \delta_{ij} \left(\sum_{l \in \text{num}} (x_{il} - c_{jl})^2 + \gamma \sum_{m \in \text{cat}} d(x_{im}, c_{jm}) \right)$$

Where:

- x_{il} : The l -th numerical feature of the i -th sample
- c_{jl} : The l -th numerical feature of the j -th cluster center
- x_{im}, c_{jm} : Corresponding categorical features of the sample and cluster center
- $d(\cdot, \cdot)$: Matching dissimilarity for categorical values (1 if different, 0 if the same)

Results We Determine the Optimal Value of k Using the Elbow Method: The Elbow Method is used to determine the optimal number of clusters (k) by plotting the within-cluster sum of squares (WCSS) against the number of clusters. The point where the rate of decrease in WCSS sharply changes (resembles an elbow) is considered the optimal value of k .

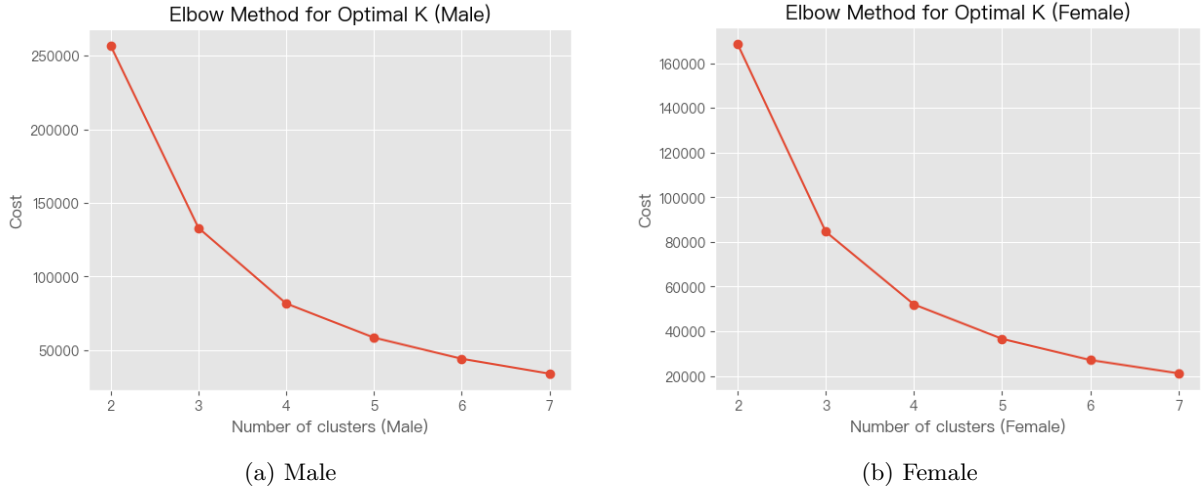


Figure 6: Determine the Optimal Value of k Using the Elbow Method

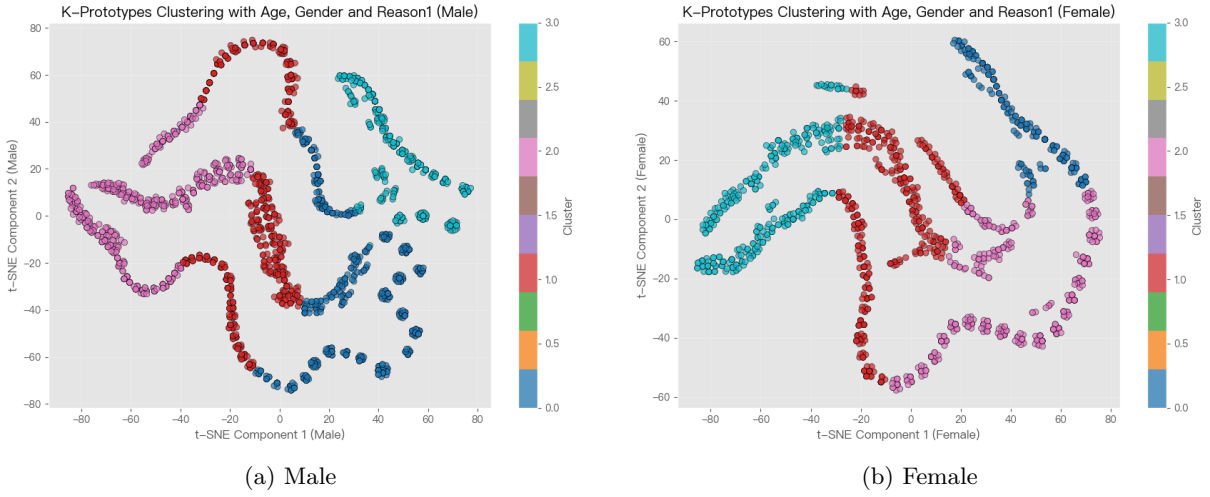


Figure 7: Cluster Results

Table 8: Male Clusters

Cluster	Age	Gender	reason1	Count
0	60.28	Male	Health	583
1	42.82	Male	Finance	651
2	24.68	Male	Health	578
3	78.67	Male	Health	360

Table 9: Female Clusters

Cluster	Age	Gender	reason1	Count
0	78.35	Female	Health	208
1	40.70	Female	Health	428
2	58.48	Female	Health	395
3	22.49	Female	Relationship	343

Analyzing the results from tables, the following insights can be drawn:

- **Health-Driven Suicides Dominate the Elderly Population:** This trend is observed in Male

Clusters 0 & 3 and Female Clusters 0 & 2, where the elderly (both males and females) are primarily driven to suicide due to health-related issues.

- **Males Are More Likely to Suicide Due to Financial Reasons:** This is seen in Male Cluster 1, where financial concerns are a dominant factor leading to suicides among males.
- **Relationship Issues Concentrated in Younger Adults:** This is evident in Female Cluster 3, where younger females are more likely to experience suicides caused by relationship problems.

Evaluation The Silhouette Coefficient is a metric used to evaluate the effectiveness of clustering, considering both the intra-cluster cohesion (tightness within a cluster) and inter-cluster separation (distance between clusters). Its value ranges from $[-1, 1]$, with values closer to 1 indicating better clustering performance. When using a custom distance for calculating the silhouette coefficient, it allows you to define a distance metric that might better reflect the specific structure or relationships within your data, beyond the standard Euclidean distance.

Table 10: Silhouette Coefficient on Gender

Category	Silhouette Coefficient
Male	0.5718
Female	0.5738

5.2 Association Rule Mining among Age, Gender, and Reason

In this section, association rule mining was conducted using the Apriori algorithm, focusing on the features **age**, **gender**, and **reason for suicide**.

Data Preprocessing

1. Selected relevant fields from the original dataset: **Gender**, **Age**, and **reason1** (main reason for suicide).
2. Removed rows with unknown **reason1** values to ensure data quality.
3. Converted the **Age** field to numeric, treating invalid values such as “不詳” as missing (NaN).
4. Grouped **Age** into meaningful ranges (e.g., 21–40, 41–60, 61+), improving analysis stability.
5. Replaced missing **Gender** values with “Unknown” to avoid issues during encoding.

Method The Apriori algorithm from `mlxtend` was used to extract frequent itemsets:

- Mined rules using key features: **Gender**, **AgeGroup**, and **Reason**.
- Dynamically set `min_support`:
`min_support = 0.05` if dataset is large (>1000 entries), else `0.1`
- Used `max_len = 3` to capture meaningful 3-element patterns.

Results Here is the results of association rules^{[11](#)}

Table 11: Top 10 Association Rules Among Age, Gender, and Reason

Index	Antecedents	Consequents	Support	Conf.	Lift
31	(AgeGroup=61+, Gender= 女)	(Reason= 健康)	0.0813	0.7989	1.7552
17	(Reason= 財政)	(Gender= 男)	0.1284	0.7910	1.2923
26	(AgeGroup=41-60, Reason= 財政)	(Gender= 男)	0.0619	0.7692	1.2567
11	(AgeGroup=61+)	(Reason= 健康)	0.2167	0.7594	1.6684
35	(Gender= 男, AgeGroup=61+)	(Reason= 健康)	0.1354	0.7375	1.6203
9	(AgeGroup=61+)	(Gender= 男)	0.1836	0.6434	1.0511
36	(AgeGroup=61+, Reason= 健康)	(Gender= 男)	0.1354	0.6248	1.0208
2	(Reason= 感情)	(AgeGroup=21-40)	0.0751	0.5428	1.9187
18	(AgeGroup=41-60, Gender= 女)	(Reason= 健康)	0.0673	0.5263	1.1563
34	(Gender= 男, Reason= 健康)	(AgeGroup=61+)	0.1354	0.5207	1.8249

Insights

- **Correlation Between Age and Health Issues:** Rules 31, 11, and 35 show a strong association between individuals aged 61+ and health-related reasons. High confidence (e.g., 0.7989) and lift (>1.6) values confirm the significance.
- **Financial Reasons and Gender Correlation:** Rules 17 and 26 indicate that financial concerns are more associated with males, revealing a gender-based pattern in financial-driven suicides.
- **Relationship Reasons and Younger Age Groups:** Rule 2 shows that emotional or relationship-related suicides are more frequent among individuals aged 21–40, with a lift close to 2, indicating a strong association.

Overall Observations These association rules provide valuable insights into how age and gender interplay with different reasons behind suicidal behavior. Patterns such as elderly individuals being more affected by health issues, or males by financial pressure, help highlight the importance of demographic-sensitive mental health interventions.

5.3 Influence of the Hong Kong Stock Market on Suicide Rates

Correlation Between Hang Seng Index and Suicide Rates This section explores the relationship between changes in the Hang Seng Index and suicide rates in Hong Kong.

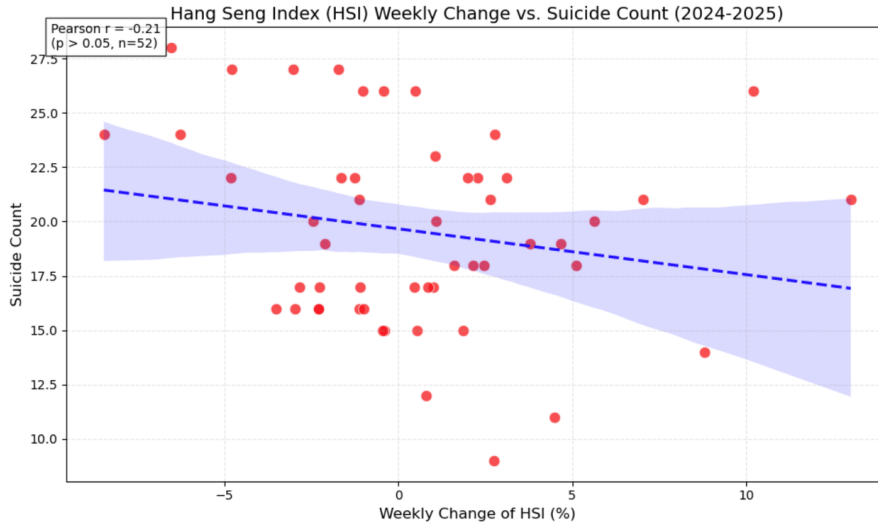


Figure 8: Hang Seng Index and Suicide Numbers in Hong Kong

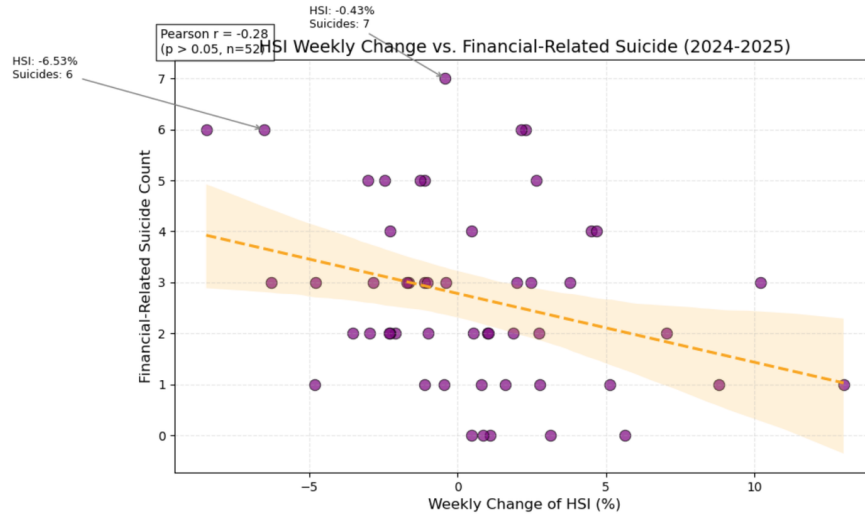


Figure 9: Hang Seng Index and Number of Suicides Due to Financial Reasons

Table 12: Pearson Correlation Between Hang Seng Index Changes and Suicide Counts

Variable Pair	Pearson Correlation Coefficient
Hang Seng Index Change (%) vs. Total Suicide Count	-0.21
Hang Seng Index Change (%) vs. Financial-Reason Suicides	-0.28

Insights

- **Total Suicide Count:** There is a weak negative correlation between the Hang Seng Index' s percentage change and the total number of suicides (correlation coefficient: -0.21). This suggests a slight tendency for suicides to increase when the stock market declines.
- **Financial-Reason Suicides:** A slightly stronger negative correlation is observed between the Hang Seng Index' s percentage change and suicides attributed to financial reasons (correlation coefficient: -0.28), indicating a more direct but still modest link.

Overall Observations These findings [Soleymani et al. \(2019\)](#) imply that stock market downturns may contribute to increased suicide risk, especially those driven by financial stress. However, the correlation remains weak, suggesting that other socioeconomic and psychological factors likely play a more significant role in influencing suicide rates [Lee et al. \(2010\)](#).

5.4 Suicide Risk Level Modeling by Population Group

This section presents the modeling process of suicide risk levels across different population groups.

Data Preprocessing To ensure the quality and consistency of data used for modeling, the following preprocessing steps were applied:

- After inspection, only a small amount of missing data was found and removed using `dropna`.
- **Age Field:**
 - Entries labeled as “Unknown” were deleted.
 - Age values were standardized.
- **State Field:**
 - Ordinal encoding was applied.
 - Similar states such as *confused*, *disoriented*, and *unconscious* were grouped under the unified category: “Impaired Consciousness”.

- The state variable was categorized into four distinct levels.
- **Binary Conversion:**
 - The fields `suicideNote`, `mental_illness`, and `emotional_illness` were converted into binary features (0 or 1).

Feature Selection Feature importance analysis revealed the following:

- `type1_encoded` and `suicideNote` are the most influential predictors.
- `emotional_illness` and `mental_illness` contribute moderately.
- `Gender` and `Age_scaled` have relatively low impact.

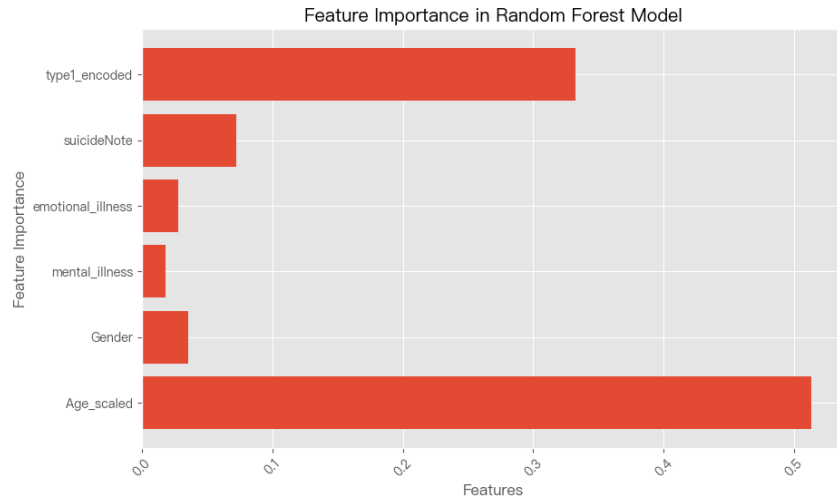


Figure 10: Feature Importance of Suicide Risk Level

Cross-Validation We used 5-fold cross-validation to evaluate model performance:

```
scores = cross_val_score(model, features, labels, cv=5, scoring='f1_weighted')
```

Logistic Regression

- F1 Score: 0.69 (± 0.01), Accuracy: 0.75, Precision: 0.67, Recall: 0.75

Table 13: Confusion Matrix: Logistic Regression

	Class 0	Class 1	Class 2	Class 3
Class 0	219	0	1021	0
Class 1	46	0	386	0
Class 2	124	0	4710	0
Class 3	8	0	22	0

Random Forest

Hyperparameter Tuning GridSearchCV was used to perform exhaustive search over specified hyperparameter values with cross-validation to optimize model performance.

- Best Parameters: `{ 'max_depth': 10, 'min_samples_split': 10, 'n_estimators': 100 }`
- F1 Score: 0.75 (± 0.02), Accuracy: 0.78, Precision: 0.75, Recall: 0.78

Table 14: Confusion Matrix: Random Forest

Class 0	Class 1	Class 2	Class 3
549	21	669	1
78	50	304	0
324	14	4496	0
17	0	13	0

ROC Curve Analysis This ROC analysis confirms the model’s strong discriminative ability in multi-class classification:

- Micro-average AUC: 0.96 (excellent overall performance)
- Macro-average AUC: 0.90 (strong performance across all classes)
- Class 3 shows near-perfect separation ($\text{AUC} = 0.98$)
- Classes 0–2 perform slightly lower but balanced ($\text{AUC} = 0.87\text{--}0.88$)

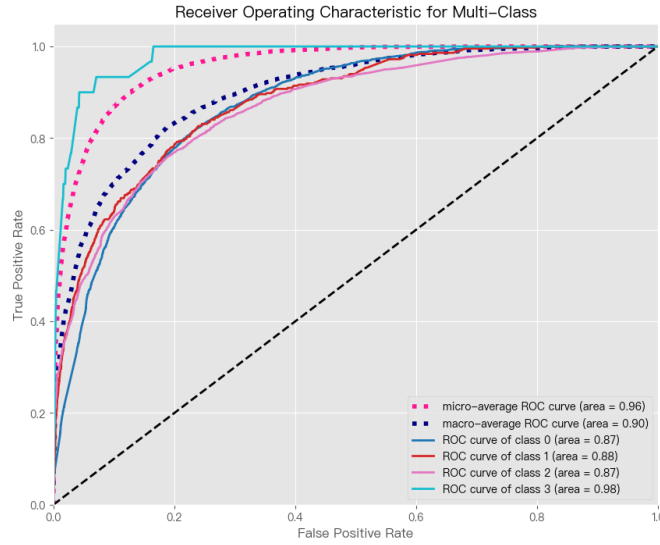


Figure 11: ROC Curve Analysis

6 Conclusion

This study presents a multi-level analysis of suicide patterns across global, East Asian, and Hong Kong contexts by integrating statistical modeling, machine learning, time-series analysis, and individual-level data mining.

At the global level, higher life expectancy is strongly associated with lower suicide rates, while factors like health spending, per capita GDP, and inequality show complex and sometimes unexpected associations. These findings emphasize the importance of viewing suicide through both public health and structural inequality lenses [Nock et al. \(2008\)](#).

In East Asia, distinct patterns emerge. South Korea shows a strong correlation between GDP fluctuations and suicide rates, especially among middle-aged adults [Chang et al. \(2013\)](#), indicating that economic volatility significantly affects mental health. In China, the unemployment rate has a more localized effect on specific age groups. Japan shows less immediate statistical linkage between macroeconomic shifts and suicide, but historical crises such as the 1997–1998 Asian Financial Crisis still reveal delayed, group-specific impacts.

In Hong Kong, suicides are spatially clustered in densely populated urban districts. Association rule mining highlights strong connections between specific locations and suicide methods (e.g., jumping from buildings, charcoal burning in cars), suggesting the need for geographically targeted interventions.

Clustering and rule mining on individual-level data reveal that elderly individuals are more affected by health-related causes, middle-aged men by financial stress, and younger women by relationship issues. Moreover, a weak but notable negative correlation between the Hang Seng Index and suicide rates—particularly for suicides due to financial reasons—indicates that economic sentiment plays a contributing role in vulnerable populations.

Through our systematic analysis of suicide data across these three levels, we uncovered multiple potential risk factors and latent patterns. However, the significance of this study extends beyond data. Ultimately, it aims to evoke deeper reflection on the value of life. We hope to express our sincere reverence for life itself and to remind every reader: life is unique, fragile, and precious—it must be protected with care and empathy.

In conclusion, suicide is a multifactorial phenomenon influenced by structural, social, economic, and personal factors. Effective prevention requires integrated approaches—combining macroeconomic monitoring, geospatial risk detection, and individual-level mental health strategies. Future research should adopt hybrid and real-time models to improve responsiveness and guide more targeted interventions across different population groups.

References

- Agrawal, R., Imieliński, T., & Swami, A. (1993). Mining association rules between sets of items in large databases. In *Acm sigmod international conference on management of data* (pp. 207–216).
- Chang, S.-S., Gunnell, D., Sterne, J. A., Lu, T.-H., & Cheng, A. T. (2013). Impact of 2008 global economic crisis on suicide: time trend study in 54 countries. *BMJ*, *347*, f5239.
- Ding, J., Chen, Y., Bressington, D., et al. (2020). Global, regional, and national burden of suicide mortality 1990 to 2016: systematic analysis for the global burden of disease study 2016. *BMJ*, *368*, 194. DOI: 10.1136/bmj.l94
- Ester, M., Kriegel, H.-P., Sander, J., & Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the 2nd international conference on knowledge discovery and data mining (kdd-96)* (pp. 226–231).
- Franklin, J. C., Ribeiro, J. D., Fox, K. R., Bentley, K. H., Kleiman, E. M., Huang, X., ... Nock, M. K. (2017). Risk factors for suicidal thoughts and behaviors: A meta-analysis of 50 years of research. *Psychological Bulletin*, *143*(2), 187–232. Retrieved from <https://doi.org/10.1037/bul0000084> DOI: 10.1037/bul0000084
- Go, T., Kim, H., & Yoon, S. e. a. (2024). The short-term effect of ambient particulate matter on suicide risk in 66 major cities and districts in south korea. *Scientific Reports*, *14*(1). DOI: 10.1038/s41598-024-52562-x
- Lee, S., Chan, K. W., Chau, A. M. L., Kwok, K. P., & Kleinman, A. (2010). Evidence for the 2008 economic crisis exacerbating depression in hong kong. *Journal of Affective Disorders*, *126*(1-2), 125–133.
- Nock, M. K., Borges, G., Bromet, E. J., Alonso, J., Angermeyer, M., Beautrais, A., & Williams, D. R. (2008). Cross-national prevalence and risk factors for suicidal ideation, plans and attempts. *The British Journal of Psychiatry*, *192*(2), 98-105.
- Soleymani, M., Ledema, B., Wong, P. W.-C., Iqbal, U., Qureshi, M. A., & Wang, Y.-Y. (2019). Assessing the impact of the economic crises in 1997 and 2008 on suicide in hong kong, taiwan and south korea. *BMC Public Health*, *19*(1), 1613.
- Taira, T., Tanaka, H., Okahisa, J., Yamaguchi, T., Ui, A., & Fukuhara, S. (2021). Association of social and economic factors with the risk of suicide in japan: A time-series analysis, 1986-2016. *Journal of Epidemiology*, *31*(5), 332–339. DOI: 10.2188/jea.JE20200171
- Tavoosi, S. (2022). *suicide-data-full-interactive-dashboard*. Kaggle Dataset. Retrieved from <https://www.kaggle.com/code/tavoosi/suicide-data-full-interactive-dashboard> (Accessed: 2025-04-24)
- Wong, W. H., Cheung, C. H., Liu, K. Y., Wong, S. T., & Yip, P. S. (2017). Stock market fluctuations and self-harm among children and adolescents in hong kong. *International Journal of Environmental Research and Public Health*, *14*(12), 1500.
- World Health Organization. (2021). Suicide worldwide in 2019: global health estimates. *World Health Organization*. Retrieved from <https://www.who.int/publications/i/item/9789240026643> (License: CC BY-NC-SA 3.0 IGO)
- World Health Organization. (2023). *Global health observatory*. <https://www.who.int/data/gho>. (Accessed: 2025-04-24)
- Xu, W., Wang, H., Wang, B., Zhang, X., Xu, Y., Chen, H., ... others (2017). Suicide rates and socioeconomic factors in eastern european countries after the collapse of the soviet union: trends between 1990 and 2008. *BMC Psychiatry*, *17*(1), 182. DOI: 10.1186/s12888-017-1358-3
- Yeong, B. (Ed.). (2025). 香港自殺報道資料庫 (*hkspd*) . <https://hkspd.siujeong.com>. (此數據截至 2025 年 4 月)

Appendix: Author Q&A

Zhixuan WANG:

When dealing with missing values, you mentioned excluding countries with more than 50% missing key indicators. Could this lead to sample selection bias? Have you considered using more complex methods such as multiple imputation to retain more data? Are rules with high confidence but low support (such as Confidence=0.54, Support=0.08) in association rules meaningful? How to balance the interpretability of rules with statistical significance?

Author response:

Thank you for these important questions. Excluding countries with substantial missing data does introduce a risk of sample selection bias. However, as our cross-country analysis relies on key socioeconomic indicators, retaining records with very limited information could impair the overall reliability of our models. While we considered advanced imputation methods, such as multiple imputation, the diversity, and complexity of the indicators across countries posed substantial methodological and practical challenges. In future iterations, we plan to explore hybrid strategies that balance coverage with data quality.

Regarding association rules: Rules with high confidence but low support—like Confidence = 0.54, Support = 0.08—can reveal interesting but less prevalent patterns in the data. However, to avoid over-interpreting rare associations, we evaluated rules using multiple measures (e.g., lift) and set minimum thresholds for both support and confidence. Balancing interpretability and statistical significance remains a challenge, and we encouraged domain-expert review alongside quantitative selection.

Zhixuan WANG & Siqiao ZHENG:

When dealing with missing values, you mentioned excluding countries with more than 50% missing key indicators. Could this lead to sample selection bias? Have you considered using more complex methods such as multiple imputation to retain more data? Are rules with high confidence but low support (such as Confidence=0.54, Support=0.08) in association rules meaningful? How to balance the interpretability of rules with statistical significance? The content is very rich, comprehensive and complete, and the theme and conclusions are very meaningful. I think it would be better to add some more detailed introduction to the method.

Author response:

Thank you very much for your suggestion. We will enrich the introduction of the method in the report, for example by including details about parameter selection and specific modeling choices to aid understanding and reproducibility.

Xin WANG:

Some data is missing, such as only seeing a map of population density in Hong Kong, but not the suicide rate for each region, making it impossible to prove the impact of population density on suicide rates.

Author response:

Thank you for your valuable observation. Considering the limited time for the presentation, we did not go through the details of each region. However, we will provide a comprehensive breakdown of suicide rates by region in the written report, including visualizations and statistical correlations, to more rigorously examine the potential impact of population density on suicide rates in Hong Kong.

Jixuan MA:

1. I have a suggestion for the "HK view individual" slide. The positions of tables and images do not correspond vertically, which is confusing.
2. Can we conduct the same analysis on the World dataset as on the HK dataset and observe if the same conclusions can be drawn?

Author response:

1. Sorry, there was a formatting issue with our PowerPoint slide. We will revise the layout in the report and future presentations to make the visual correspondence clearer.
2. Thank you for your question. From a data availability perspective, Hong Kong provides highly detailed and publicly accessible data, including individual-level information such as age, gender, method,

and location of suicide. This allows for more in-depth analysis. In contrast, most global datasets only offer aggregated statistics—like overall suicide rates or breakdowns by age or gender—without detailed individual-level data. So even if we wanted to apply the same analysis globally, the lack of detailed data makes it difficult. From a research standpoint, our project intentionally separates macro-level (e.g., global data) and micro-level (e.g., Hong Kong data) analyses. These datasets differ in structure and serve different analytical purposes, forming a complementary framework rather than being directly comparable.

Yuxiao MA:

This is an extremely important and sensitive topic that deserves significant attention and consideration. I appreciate the comprehensive survey conducted using a variety of data mining techniques. To further enrich the analysis of individual behavioral patterns related to suicide rates, it might be valuable to incorporate additional features, such as education level and the prevalence or accessibility of mental health care services. Exploring these factors could provide deeper insights and help paint a more complete picture of the underlying issues.

Author response:

That would indeed allow for a more comprehensive analysis of the issue. However, due to the limited scope of the project, we were not able to explore these aspects in depth. We will consider adjusting the content accordingly in future iterations.

Jiajun FANG:

The PPT is done great! The content is full of practical knowledge, but it is not boring at all. The color matching and layout look particularly comfortable. The speech is also super captivating, with a natural and infectious tone. Especially the analysis in the random forest section, I really gained a lot after listening!

Author response:

Thank you so much, Jiajun! Appreciate your support!

Pei Ci YEE:

The group chose an interesting topic and I learned a new cluster method which is the K-prototype which combines the strength of K-means and K-modes through watching this presentation. I have one question. Do you consider using the same method for all of the datasets?

Author response:

Thank you for your comment. However, from a research perspective, our project itself distinguishes between macro and micro levels of analysis. First, different datasets are often used to address different tasks, which naturally leads to the use of different methods. Hong Kong provides highly accessible data, including individual-level details such as age, gender, method of suicide, and location, which enables more in-depth analysis. In contrast, most other countries only release aggregated statistics, lacking the micro-level information needed for such detailed studies.

Ke FAN:

It might be helpful to include a brief explanation of the data processing in the “East Asia View” section.

Author response:

We will definitely include that in the report, along with a detailed explanation of the methods used.